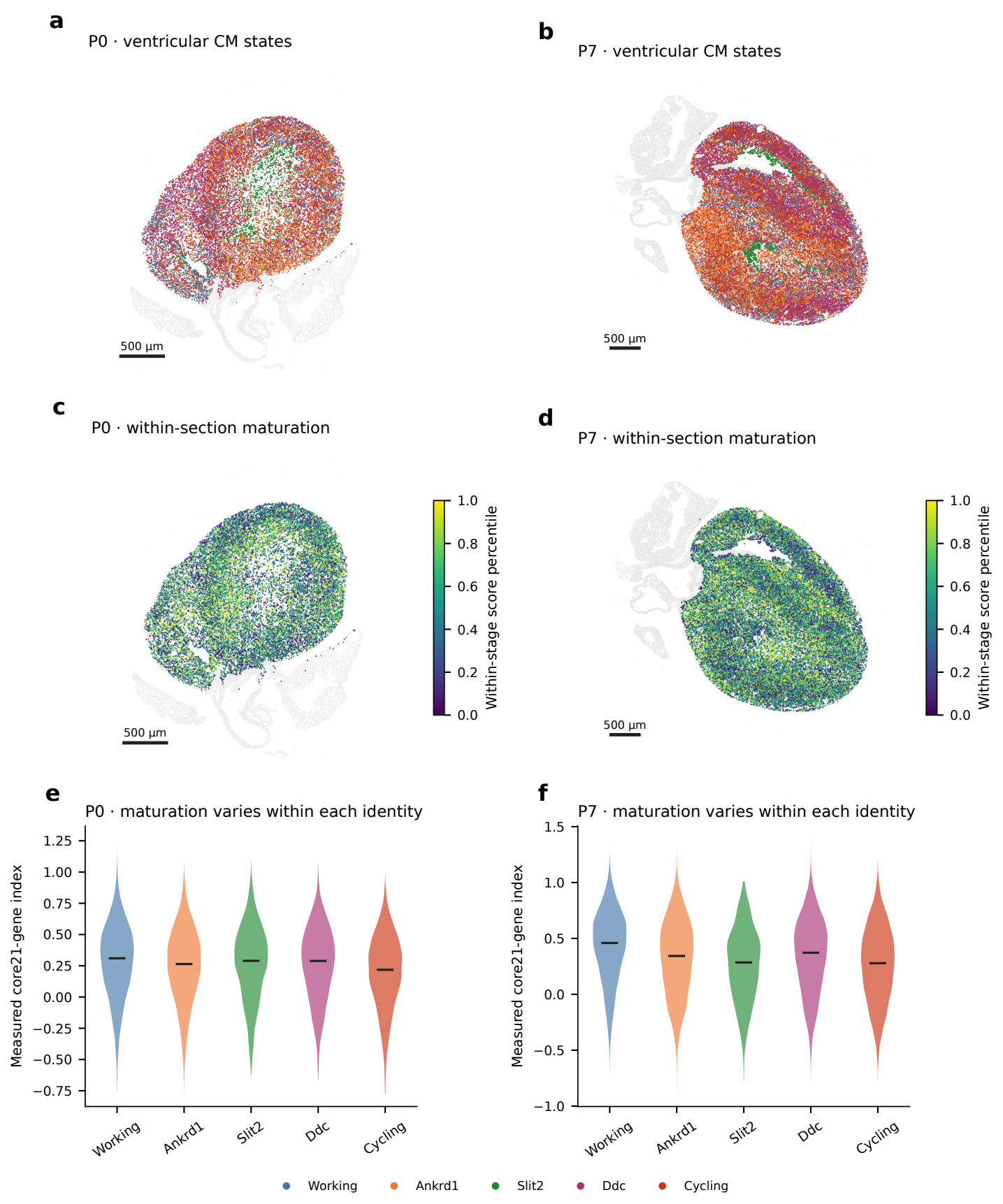
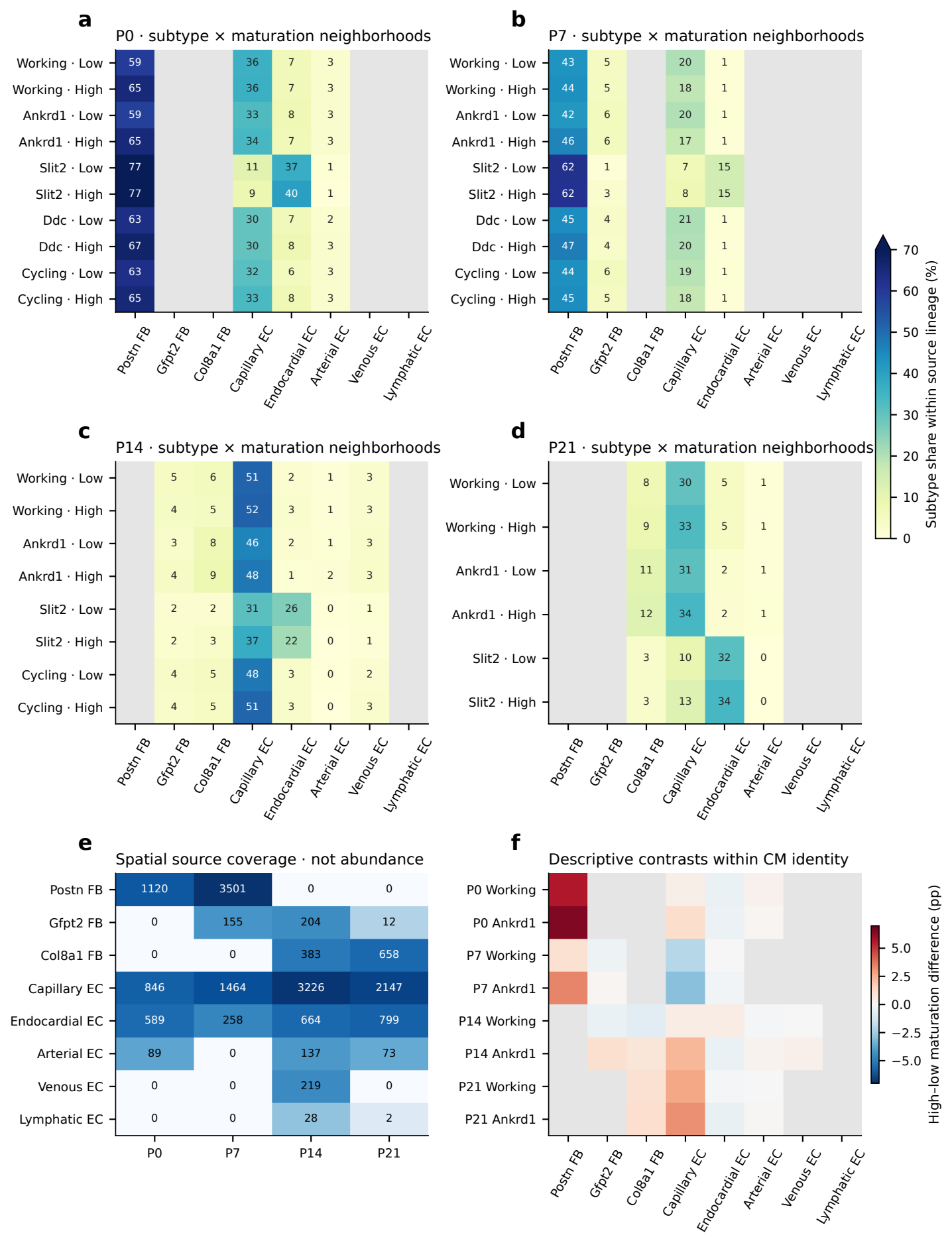
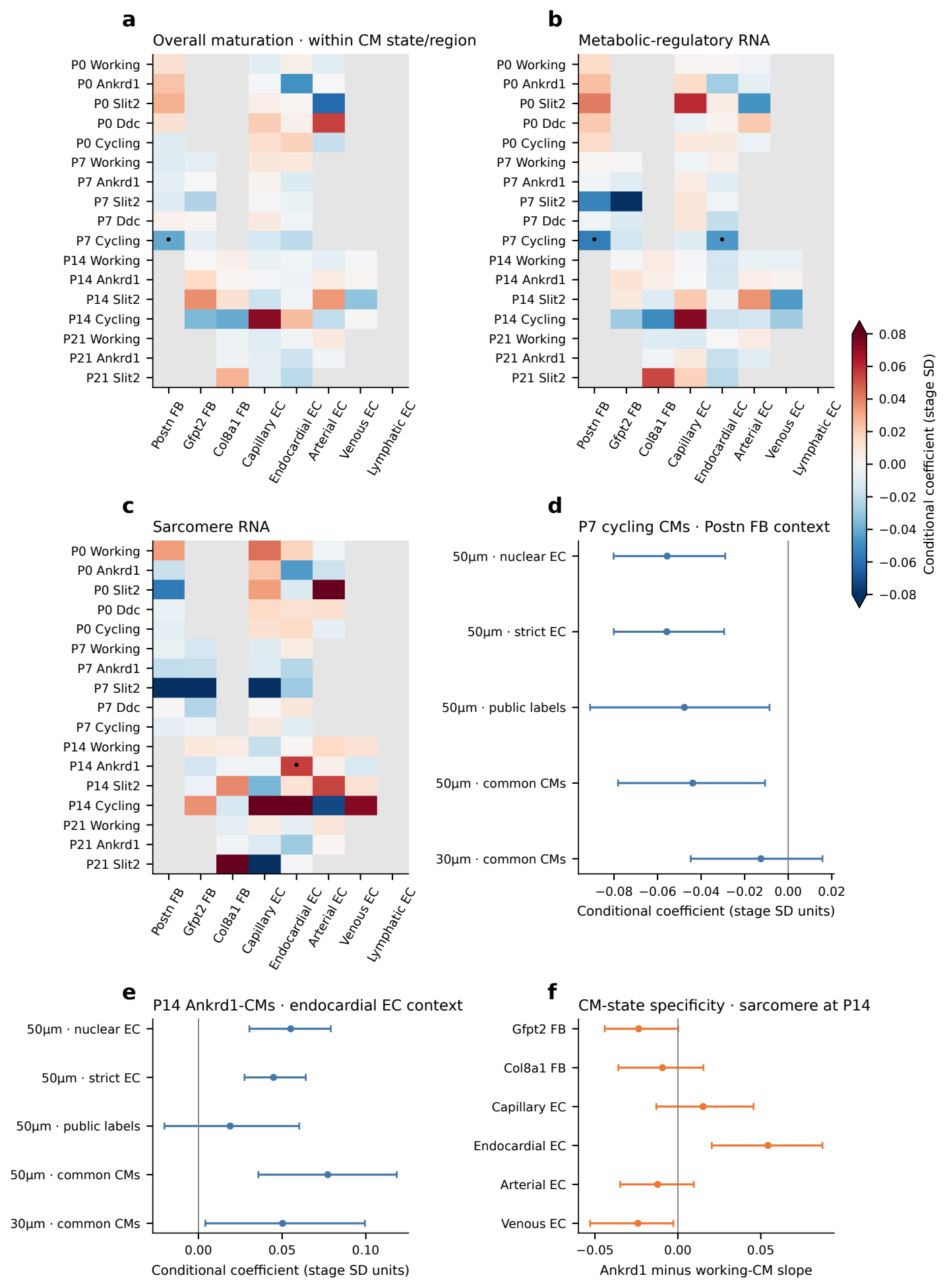
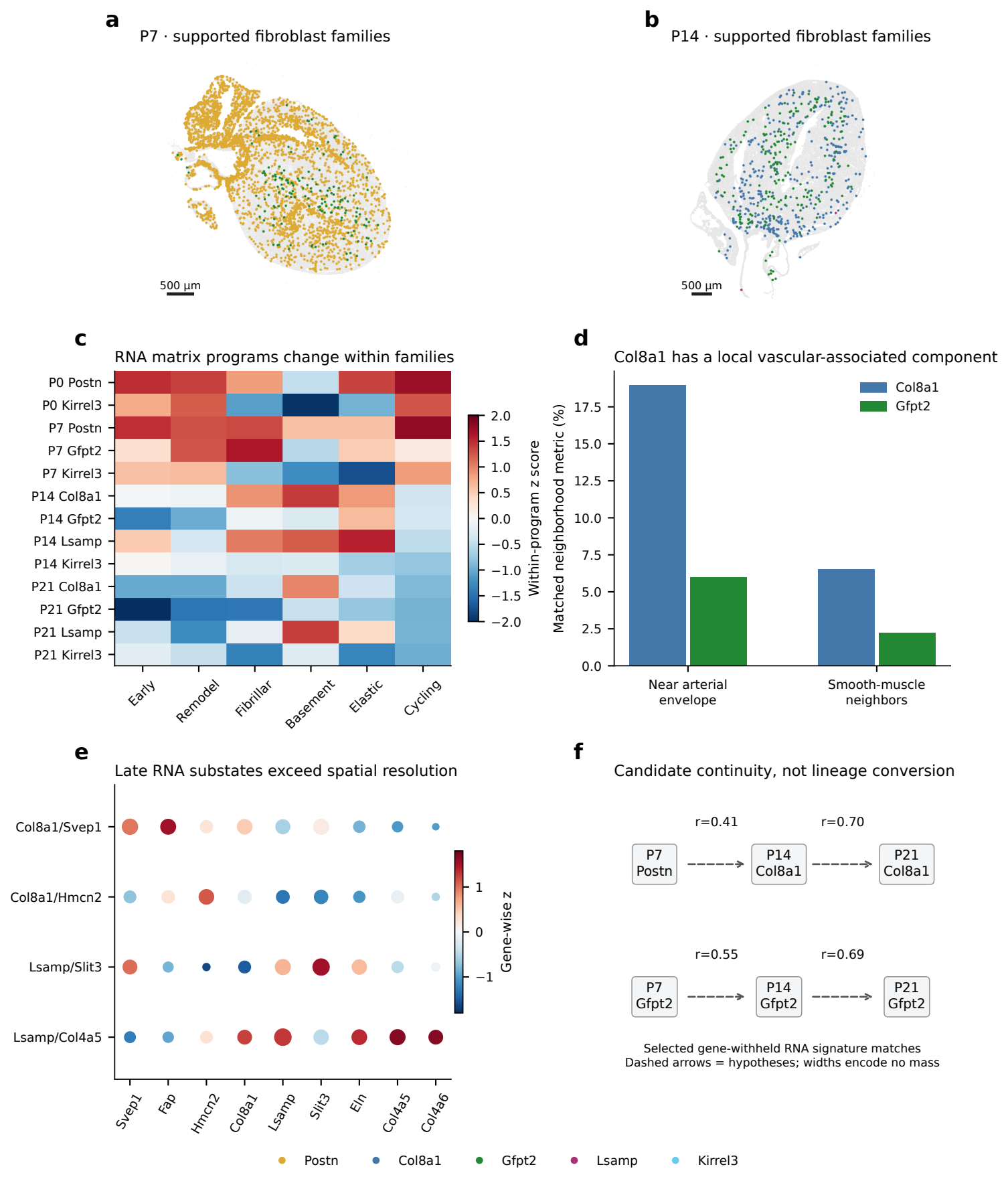
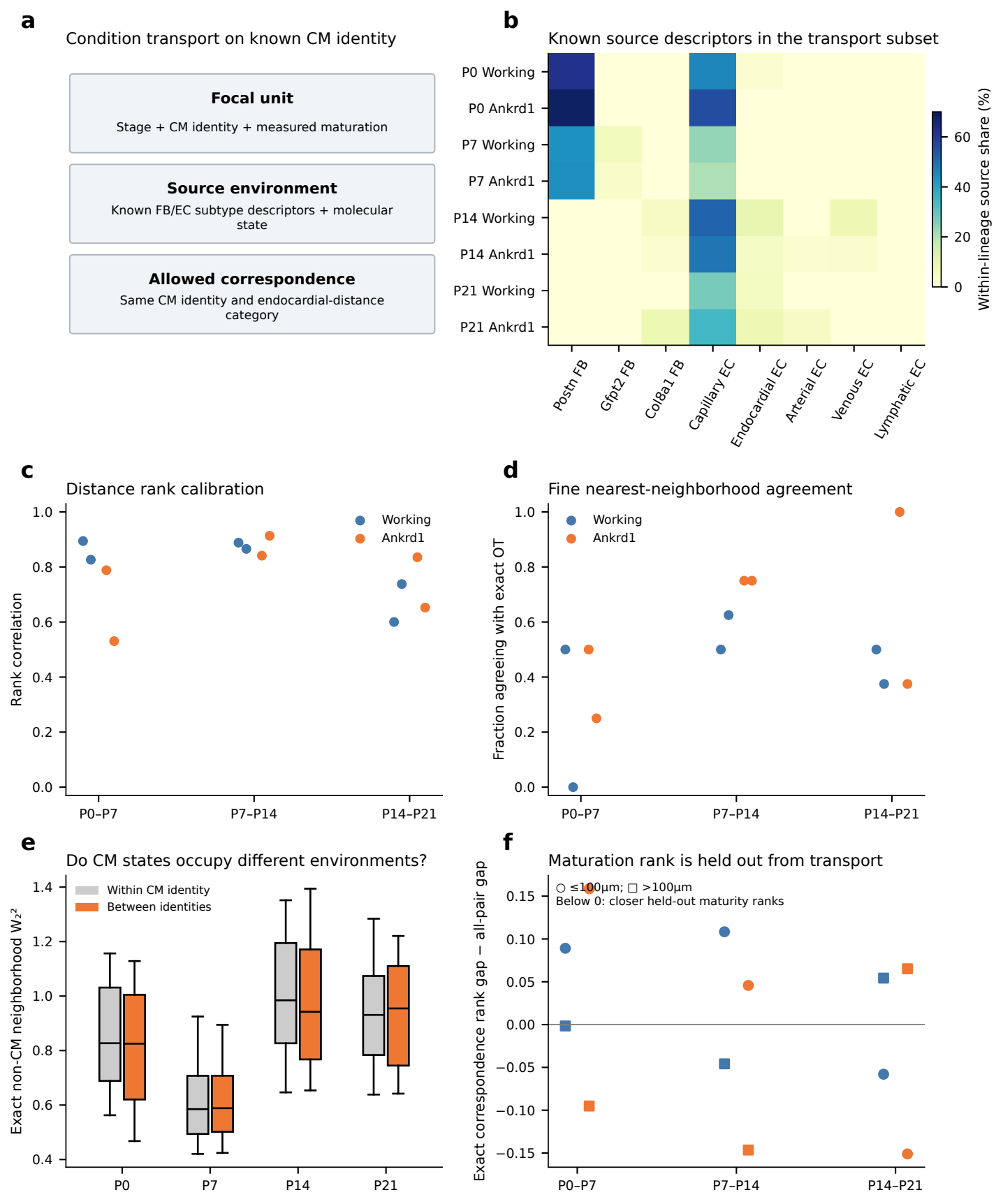

a
P0 · ventricular CM states
500 µm
b
P7 · ventricular CM states
500 µm
c
P0 · within-section maturation
500 µm
1.0
0.8
0.6
0.4
0.2
0.0
Within-stage score percentile
d
P7 · within-section maturation
500 µm
1.0
0.8
0.6
0.4
0.2
0.0
Within-stage score percentile
e
P0 · maturation varies within each identity
Measured core21-gene index
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
Working
Ankrd1
Slit2
Ddc
Cycling
f
P7 · maturation varies within each identity
Measured core21-gene index
1.5
1.0
0.5
0.0
−0.5
−1.0
Working
Ankrd1
Slit2
Ddc
Cycling
Working
Ankrd1
Slit2
Ddc
Cycling


a
P0 · subtype × maturation neighborhoods
Working · Low 59 36 7 3
Working · High 65 36 7 3
Ankrd1 · Low 59 33 8 3
Ankrd1 · High 65 34 7 3
Slit2 · Low 77 11 37 1
Slit2 · High 77 9 40 1
Ddc · Low 63 30 7 2
Ddc · High 67 30 8 3
Cycling · Low 63 32 6 3
Cycling · High 65 33 8 3
Postn FB
Gfpt2 FB
Col8a1 FB
Capillary EC
Endocardial EC
Arterial EC
Venous EC
Lymphatic EC
b
P7 · subtype × maturation neighborhoods
Working · Low 43 5 20 1
Working · High 44 5 18 1
Ankrd1 · Low 42 6 20 1
Ankrd1 · High 46 6 17 1
Slit2 · Low 62 1 7 15
Slit2 · High 62 3 8 15
Ddc · Low 45 4 21 1
Ddc · High 47 4 20 1
Cycling · Low 44 6 19 1
Cycling · High 45 5 18 1
Postn FB
Gfpt2 FB
Col8a1 FB
Capillary EC
Endocardial EC
Arterial EC
Venous EC
Lymphatic EC
Subtype share within source lineage (%)
70
60
50
40
30
20
10
0
c
P14 · subtype × maturation neighborhoods
Working · Low 5 6 51 2 1 3
Working · High 4 5 52 3 1 3
Ankrd1 · Low 3 8 46 2 1 3
Ankrd1 · High 4 9 48 1 2 3
Slit2 · Low 2 2 31 26 0 1
Slit2 · High 2 3 37 22 0 1
Cycling · Low 4 5 48 3 0 2
Cycling · High 4 5 51 3 0 3
Postn FB
Gfpt2 FB
Col8a1 FB
Capillary EC
Endocardial EC
Arterial EC
Venous EC
Lymphatic EC
d
P21 · subtype × maturation neighborhoods
Working · Low 8 30 5 1
Working · High 9 33 5 1
Ankrd1 · Low 11 31 2 1
Ankrd1 · High 12 34 2 1
Slit2 · Low 3 10 32 0
Slit2 · High 3 13 34 0
Postn FB
Gfpt2 FB
Col8a1 FB
Capillary EC
Endocardial EC
Arterial EC
Venous EC
Lymphatic EC
e
Spatial source coverage · not abundance
Postn FB 1120 3501 0 0
Gfpt2 FB 0 155 204 12
Col8a1 FB 0 0 383 658
Capillary EC 846 1464 3226 2147
Endocardial EC 589 258 664 799
Arterial EC 89 0 137 73
Venous EC 0 0 219 0
Lymphatic EC 0 0 28 2
P0
P7
P14
P21
f
Descriptive contrasts within CM identity
P0 Working
P0 Ankrd1
P7 Working
P7 Ankrd1
P14 Working
P14 Ankrd1
P21 Working
P21 Ankrd1
Postn FB
Gfpt2 FB
Col8a1 FB
Capillary EC
Endocardial EC
Arterial EC
Venous EC
Lymphatic EC
High–low maturation difference (pp)
5.0
2.5
0.0
−2.5
−5.0


a
Overall maturation · within CM state/region
b
Metabolic-regulatory RNA
c
Sarcomere RNA
d
P7 cycling CMs · Postn FB context
e
P14 Ankrd1-CMs · endocardial EC context
f
CM-state specificity · sarcomere at P14
P0 Working
P0 Ankrd1
P0 Slit2
P0 Ddc
P0 Cycling
P7 Working
P7 Ankrd1
P7 Slit2
P7 Ddc
P7 Cycling
P14 Working
P14 Ankrd1
P14 Slit2
P14 Cycling
P21 Working
P21 Ankrd1
P21 Slit2
Postn FB
Gfpt2 FB
Col8a1 FB
Capillary EC
Endocardial EC
Arterial EC
Venous EC
Lymphatic EC
Conditional coefficient (stage SD)
0.08
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08
50µm · nuclear EC
50µm · strict EC
50µm · public labels
50µm · common CMs
30µm · common CMs
−0.08 −0.06 −0.04 −0.02 0.00 0.02
Conditional coefficient (stage SD units)
0.00 0.05 0.10
Conditional coefficient (stage SD units)
−0.05 0.00 0.05
Ankrd1 minus working-CM slope


a
P7 · supported fibroblast families
500 µm
b
P14 · supported fibroblast families
500 µm
c
RNA matrix programs change within families
P0 Postn
P0 Kirrel3
P7 Postn
P7 Gfpt2
P7 Kirrel3
P14 Col8a1
P14 Gfpt2
P14 Lsamp
P14 Kirrel3
P21 Col8a1
P21 Gfpt2
P21 Lsamp
P21 Kirrel3
Early
Remodel
Fibrillar
Basement
Elastic
Cycling
Within-program z score
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
d
Col8a1 has a local vascular-associated component
Col8a1
Gfpt2
Matched neighborhood metric (%)
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Near arterial envelope
Smooth-muscle neighbors
e
Late RNA substates exceed spatial resolution
Col8a1/Svep1
Col8a1/Hmcn2
Lsamp/Slit3
Lsamp/Col4a5
Svep1
Fap
Hmcn2
Col8a1
Lsamp
Slit3
Eln
Col4a5
Col4a6
Gene-wise z
1
0
−1
f
Candidate continuity, not lineage conversion
r=0.41
r=0.70
P7 Postn
P14 Col8a1
P21 Col8a1
r=0.55
r=0.69
P7 Gfpt2
P14 Gfpt2
P21 Gfpt2
Selected gene-withheld RNA signature matches
Dashed arrows = hypotheses; widths encode no mass
Postn
Col8a1
Gfpt2
Lsamp
Kirrel3


a
Condition transport on known CM identity
Focal unit
Stage + CM identity + measured maturation
Source environment
Known FB/EC subtype descriptors + molecular state
Allowed correspondence
Same CM identity and endocardial-distance category
b
Known source descriptors in the transport subset
P0 Working
P0 Ankrd1
P7 Working
P7 Ankrd1
P14 Working
P14 Ankrd1
P21 Working
P21 Ankrd1
Postn FB
Gfpt2 FB
Col8a1 FB
Capillary EC
Endocardial EC
Arterial EC
Venous EC
Lymphatic EC
Within-lineage source share (%)
0
20
40
60
c
Distance rank calibration
Working
Ankrd1
Rank correlation
P0–P7
P7–P14
P14–P21
d
Fine nearest-neighborhood agreement
Fraction agreeing with exact OT
e
Do CM states occupy different environments?
Within CM identity
Between identities
Exact non-CM neighborhood W₂²
P0
P7
P14
P21
f
Maturation rank is held out from transport
○ ≤100µm; □ >100µm
Below 0: closer held-out maturity ranks
Exact correspondence rank gap − all-pair gap

a

The analysis unit is a subtype-defined developmental neighborhood

CM identity

Ankrd1; working; Slit2; Ddc; cycling

Maturation within stage

Continuous21-gene score and separate functional-RNA components

Non-CM subtypes

Postn / Gfpt2 / Col8a1 FBs; capillary / endocardial / arterial ECs

Broad lineage context is background; it is not a limit on subtype-defined analyses

b

Component-specific candidates from the observed data

P7 · cycling CM + Postn FB

Lower maturation/metabolic-regulatory RNA; annotation and radius sensitivity remain

P14 · Ankrd1-CM + endocardial EC

Higher sarcomere RNA in local models; not a uniform increase in all maturation components

c

Choose subtype-matched functional and lineage experiments

Fibroblast comparison

Postn versus Gfpt2 context around early cycling CMs; assess metabolism, matrix and completed division

EC × CM identity

Endocardial-like versus capillary ECs around Ankrd1 versus working CMs; sarcomeres, calcium and force

State continuity

Trace validated early FB labels into later matrix-specialized states; do not treat OT as ancestry